

Bachelor's Thesis

Voice-Cloning-Based Social Engineering

Voice-Cloning-basiertes Social Engineering

Submitted by **Abdul Samed Tuna**
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supervised by: Dr. Christian Steineder

Kurzfassung

Voice Cloning ermöglicht es, charakteristische Merkmale einer bekannten Stimme synthetisch nachzubilden. Eine überzeugende synthetische Stimme stellt jedoch für sich genommen noch keinen erfolgreichen Social-Engineering-Angriff dar. Diese Arbeit untersucht, wie geklonte Sprache bei telefonbasierter Identitätsvortäuschung sicherheitsrelevant werden kann, wenn die wahrgenommene Identität des Anrufers eine geschützte Autorisierungsentscheidung beeinflusst.

Hier für wird ein anrufspezifisches sozio-technisches Bedrohungsmodell entwickelt, das Identitätsdarstellung, menschliche Autorisierung und die Ausführung eines geschützten Prozesses durch drei eigens entwickelte Grenzen (B1-B3) voneinander trennt. Ein Angriffsbaum beschreibt die logischen Bedingungen für einen sicherheitsrelevanten Erfolg, während ein siebenstufiger Ablauf deren zeitliche Abfolge darstellt. Zusätzlich unterscheidet das Modell zwischen Identitätserfolg, Interaktionserfolg und sicherheitsrelevantem Erfolg.

Zwei konstruierte Szenarien untersuchen unterschiedliche soziale Einflussmechanismen. Im ersten Szenario stehen hierarchische Autorität und Dringlichkeit im Vordergrund, im zweiten begrenzte Vertrautheit, emotionale Belastung und Serviceorientierung. Beide Szenarien zeigen, wie unterschiedliche soziale Bedingungen zu derselben Autorisierungsschwäche beitragen können.

Eine ergänzende lokale deutschsprachige technische Validierung mit XTTS-v2, RVC und Seed-VC untersucht Inhaltserhalt, Sprecherähnlichkeit auf Basis von Embeddings, Verarbeitungsgeschwindigkeit und wahrnehmungsbezogene Eigenschaften. Die Ergebnisse zeigen, dass Natürlichkeit, sprachliche Korrektheit, Embedding-Ähnlichkeit, wahrgenommene Zielsprecherähnlichkeit und situationsgerechte Sprechweise deutlich voneinander abweichen können und daher nicht als gleichwertige Indikatoren für Angriffsfähigkeit behandelt werden sollten. Das zentrale Ergebnis ist, dass Voice Cloning in erster Linie die Glaubwürdigkeit einer Identitätsbehauptung verstärkt; ein sicherheitsrelevanter Erfolg entsteht erst dann, wenn diese anruferbasierte Glaubwürdigkeit die unabhängig erforderlichen Autorisierungsnachweise für eine geschützte Handlung beeinflusst oder ersetzt.

Abstract

Voice cloning can reproduce characteristics of a known speaker, but a convincing synthetic voice does not by itself constitute a successful social-engineering attack. This thesis examines how cloned speech can become security-relevant in call-based impersonation when perceived caller identity influences a protected authorization decision.

A call-specific socio-technical threat model separates identity presentation, human authorization, and protected-process execution through three author-developed boundaries (B1-B3). An attack tree identifies the logical conditions required for security impact, while a seven-stage lifecycle describes their temporal progression. The model further distinguishes identity success, interaction success, and security-impact success.

Two constructed scenarios examine different social influence mechanisms. The first focuses on executive authority and urgency, while the second combines limited familiarity, distress, and service expectations. Both scenarios show how different social conditions can contribute to the same authorization weakness.

A supplementary local German-language validation using XTTS-v2, RVC, and Seed-VC evaluates content preservation, speaker-embedding similarity, processing speed, and perceptual characteristics. The results show that naturalness, linguistic correctness, embedding similarity, target-speaker resemblance, and scenario-appropriate delivery can diverge substantially and should not be treated as equivalent indicators of attack capability. The central finding is that voice cloning primarily strengthens an identity claim; security impact occurs only when caller-derived legitimacy is allowed to influence or replace the independent authorization evidence required for a protected action.

List of Abbreviations

AI	Artificial Intelligence
ASR	Automatic Speech Recognition
CER	Character Error Rate
DM	Decision-Maker
II	Impersonated Identity
OX	Organization X
RTF	Real-Time Factor
RVC	Retrieval-based Voice Conversion
TA	Threat Actor
TTS	Text-to-Speech
VC	Voice Conversion
WER	Word Error Rate
STS	Speech-to-Speech

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1. Introduction

1.1. Context and Problem Statement

Voice cloning changes an important assumption in telephone communication: a familiar voice can no longer be treated as reliable evidence that the corresponding person is speaking. Current synthetic-speech systems can reproduce characteristics of target speakers closely enough to create substantial identity confusion, while recent research on AI-enabled vishing shows that persuasive context remains at least as important as voice realism alone [1], [2]. The resulting threat is therefore not simply an audio-generation problem. It is a socio-technical security problem in which synthetic speech can strengthen a false identity claim during an interaction with a person who has legitimate decision authority.

The security-relevant transition occurs when apparent caller identity influences a protected decision. An employee may recognize what appears to be an executive, customer, or other known person without that recognition constituting the authorization evidence required for a payment, transaction change, or disclosure. Voice cloning becomes consequential when this distinction is weakened and caller-derived legitimacy is allowed to substitute for an independent authorization requirement. Existing research addresses synthetic-speech perception, vishing, persuasion, and social-engineering susceptibility, but these strands do not by themselves explain how an identity signal progresses through a live call toward a protected organizational outcome [3]. This thesis analyzes that transition explicitly.

1.2. Research Question and Objectives

The thesis addresses the following research question:

How does voice cloning alter perceived caller legitimacy in call-based social engineering, and under which technical, contextual, and human conditions can that identity signal influence a protected authorization decision?

The analysis is guided by four supporting questions:

1. Which actors, assets, adversary goals, assumptions, and trust boundaries characterize voice-cloning-based call impersonation?
2. Which prerequisites and stages connect target selection and reference-audio acquisition to identity presentation, interaction, and security impact?
3. How do contextual alignment, authority, familiarity, distress, timing, and suspicion affect the transition from perceived identity to a protected decision?
4. Which technical and procedural conditions enable or interrupt this transition?

The objective is not to estimate how frequently such attacks succeed, but to identify the conditions that must align for synthetic caller identity to become security-relevant.

1.3. Scope and Contribution

The thesis focuses on human-operated telephone impersonation in which an attacker presents as a specific person already known to the Decision-Maker. Both text-to-speech voice cloning and speech-to-speech voice conversion are considered as possible means of producing target-like speech. The protected outcomes considered are organizational or financial actions such as payment approval, transaction reauthorization, or disclosure of protected information. Autonomous conversational agents, video deepfakes, direct system compromise, attacks against voice-biometric authentication, and operational caller-ID manipulation are outside the scope.

The primary contribution is a call-specific analytical model that separates three security-relevant transitions: identity presentation, human authorization, and the protected organizational process. These are represented by the author-developed B1–B3 boundary model. An attack tree expresses the logical conditions required for security impact, while a seven-stage lifecycle orders the same problem temporally. The analysis further distinguishes *identity success*, *interaction success*, and *security-impact success*, preventing a convincing voice or sustained conversation from being treated as equivalent to successful compromise.

The model is subsequently applied to two contrasting scenarios and supported by a supplementary technical validation that examines selected practical boundary conditions of locally tested TTS and voice-conversion systems.

1.4. Thesis Structure

Chapter 2 introduces the necessary foundations in social engineering, vishing, human judgment, voice cloning, and authorization. Chapter 3 reviews related work and identifies

the research gap. Chapter 4 defines the methodology and provenance of the analytical framework. Chapter 5 develops the boundary model, attack tree, and seven-stage lifecycle. Chapter 6 applies the framework to the two constructed scenarios. Chapter 7 presents the exploratory technical validation. Chapter 8 integrates the findings, discusses their security implications and limitations, and answers the research question.

2. Background and Foundations

2.1. Social Engineering and Vishing

Social engineering attempts to influence people into disclosing information or performing actions that conflict with security objectives. The adversary does not necessarily defeat a technical control directly; instead, a person with legitimate access or decision authority is induced to use that authority in an unintended way. Vishing applies this mechanism through spoken communication. Telephone interaction is particularly relevant because identity, context, and paralinguistic cues must be interpreted while the conversation continues, making vishing a socio-technical problem rather than a purely technical or psychological one [3].

Call-based impersonation adds a specific identity claim to this interaction. The recipient evaluates not only whether a request is plausible but also whether the caller appears to be the person they claim to be. Voice cloning changes the evidence available for that judgment by allowing characteristics associated with a target speaker to be reproduced independently of that speaker. The resulting security problem therefore concerns the interaction between synthetic identity presentation, human judgment, and the process governing the requested action.

2.2. Caller Legitimacy, Suspicion, and Influence

Truth-Default Theory provides the primary behavioral foundation used in this thesis. Applied to vishing, it explains how a caller can initially be treated as legitimate until sufficiently diagnostic information creates suspicion [4]. Sensitive requests, contextual inconsistencies, unexpected behavior, or procedural requirements can therefore change the recipient's assessment during the call. Initial acceptance of a caller and continued accep-

tance through a decision-relevant interaction should consequently be treated as separate states.

Authority, urgency, and distress can modify this judgment but are not deterministic causes of compliance. Authority is recurrent in observed vishing [5], while experimental cybersecurity research indicates that increased time pressure can reduce detection performance without establishing that urgency necessarily produces unsafe decisions [6]. Distress operates through a different mechanism: experimental evidence shows that empathy toward another person’s suffering can bias decisions toward prosocial action [7]. These findings support treating authority, urgency, and distress as contextual influences on verification and decision-making rather than as substitutes for evidence that a request is authorized.

2.3. Voice Cloning and Call Interaction

Voice cloning refers here to generating or transforming speech so that it reproduces characteristics associated with a target speaker. In text-to-speech (TTS), arbitrary text is synthesized using target-speaker conditioning. In voice conversion, source speech supplies the linguistic content and performance while speaker characteristics are transformed toward a target. The approaches therefore create different operational trade-offs: TTS provides direct control over wording, whereas voice conversion can preserve source timing and prosodic performance more directly. Comparative work confirms that TTS and voice-conversion performance depends on the implementation and reference material rather than on a single general notion of synthesis quality [8].

Speaker identity is likewise not reducible to timbre alone. Speech rate, rhythm, pitch behavior, intonation, and other prosodic characteristics contribute to how synthetic speech is perceived [9]. This distinction is important because linguistic correctness, naturalness, target-speaker resemblance, and emotional delivery are different properties. A sample may be clear and natural without convincingly reproducing a particular speaker or the delivery expected in a specific situation. Familiarity further affects this interpretation because a listener who knows the claimed speaker has prior expectations against which the incoming voice can be compared [10]; its empirical implications are examined in Chapter 3.

Call interaction also introduces a timing constraint that isolated audio generation does not capture. Processing faster than the duration of the generated audio does not imply an equally fast conversation: understanding the incoming turn, deciding on a response, preparing input, synthesis or conversion, and playback all contribute to the interaction. Telecommunications latency can alter normal speaker transitions and increase gaps or overlap between turns [11]. Technical throughput and conversational latency are therefore treated separately in the later technical validation.

2.4. Identity and Authorization Evidence

A caller’s apparent identity and the authorization of a requested action are distinct security questions. A recognizable voice may support the hypothesis that a particular person is speaking, but it does not establish that a payment, disclosure, transaction change, or other protected action is authorized. Current NIST guidance reinforces this distinction in formal authentication contexts by disallowing voice-based biometric comparison as an authentication mechanism [12]. This guidance concerns formal biometric authentication rather than informal human recognition, but it demonstrates why speaker-related characteristics should not be treated as sufficient authentication evidence.

More generally, established security design principles require sensitive actions to remain subject to explicit authorization checks rather than relying on a single unverified condition [13]. For the present thesis, this provides the foundation for separating *identity evidence* from *authorization evidence*: synthetic speech can influence the former, while a protected action should depend on independently required evidence. Chapter 5 develops this distinction into the call-specific threat model.

3. Related Work

Research relevant to voice-cloning-based social engineering spans speaker perception, vishing, persuasion, and security research. These areas establish important parts of the threat separately, but they answer different questions. Voice-cloning studies examine whether synthetic speech reproduces characteristics of a target speaker; vishing research examines how callers establish and lose credibility; and social-engineering research examines how contextual and cognitive factors affect responses to deceptive requests. This chapter therefore focuses on work that directly informs the transition from synthetic identity presentation to a security-relevant human decision.

3.1. Voice Cloning, Speaker Identity, and Familiarity

Barrington, Cooper, and Farid directly distinguish between perceived speaker identity and detection of synthetic speech [1]. Using contemporary voice clones across a large set of speakers, they found that participants frequently judged a cloned voice and its genuine

counterpart to have the same identity. In the identity-matching task, the median rate of “same identity” judgments for genuine–clone pairs was 83.3%. In a separate naturalness task, participants correctly classified cloned speech as AI-generated only 60.8% of the time. The distinction between these tasks is important: uncertainty about whether speech is synthetic does not imply equivalent uncertainty about who appears to be speaking. Their experiment nevertheless isolates auditory perception from the broader interaction; participants were not making decisions for a known caller inside a social-engineering scenario.

Familiarity changes this problem. Rosi, Soopramanien, and McGettigan compared evaluations of cloned and genuine voices by listeners who knew the speaker and by unfamiliar listeners [10]. Familiar listeners were more sensitive to the accuracy and authenticity of cloning, while unfamiliar listeners sometimes evaluated synthetic versions more positively than the corresponding recordings. Within familiar listeners, perceived similarity between the clone and the known speaker predicted social evaluations of the voice. These findings do not support a simple assumption that familiarity necessarily makes a recipient more vulnerable. Familiarity can strengthen an identity hypothesis when a clone reproduces expected characteristics, but it can also provide richer speaker-specific expectations against which inconsistencies can be noticed.

Together, these studies establish two constraints for call-based impersonation. First, perceived target identity and recognition of synthetic speech are non-equivalent judgments. Second, the value of reproducing a specific voice depends partly on the listener’s relationship to that voice. Neither study, however, examines what occurs after a caller is provisionally accepted as the claimed person. In particular, they do not test whether perceived identity is subsequently treated as sufficient evidence for a protected action.

3.2. Vishing, Persuasion, and AI-Generated Calls

Vishing research provides evidence about the interaction surrounding the identity claim. Jones et al. analyzed 86 documented examples of vishing and coded the persuasion principles used by social engineers [5]. Authority, social proof, and distraction were among the most prevalent mechanisms. The study provides direct support for treating authority as a recurring feature of vishing, but its corpus design identifies which mechanisms occur rather than establishing how strongly each mechanism causes compliance. This distinction is important for the executive scenario developed later in the thesis: authority is treated as an influence mechanism whose effect depends on the surrounding interaction and authorization process, not as a deterministic cause of action.

Armstrong, Jones, and Siami Namin examine the recipient side of the interaction through Truth-Default Theory [4]. Participants evaluated eight real-world vishing conversations at

different stages. Callers were initially perceived as honest, but perceived honesty declined as diagnostic information accumulated. Requests for highly sensitive information produced a further decline, whereas seemingly innocuous requests did not produce the same effect. The study therefore provides an interaction-level explanation for why initial legitimacy and later suspicion should be analyzed separately. Its scope is nevertheless limited for the present problem: participants evaluated descriptions of existing vishing conversations, and the study neither manipulated cloned voices nor measured authorization of a protected organizational action.

Recent work by Heiding et al. brings synthetic speech and phishing scenarios substantially closer together [2]. In a large-scale study with 4,100 participants, multiple contemporary AI voice systems and human baselines were evaluated across several scam scenarios. Caller persuasiveness was the strongest predictor of stated compliance, while human-likeness alone did not determine the result. The relative-in-distress scenario produced the highest reported susceptibility, demonstrating that social framing can remain important even when technically capable synthetic speech is available. This is strong evidence against reducing AI-enabled vishing to an acoustic-realism problem. At the same time, the outcome measured whether participants reported that they would or might comply with a request; it did not observe an employee crossing a real authorization boundary or bypassing a required organizational control.

These studies therefore address complementary parts of the same interaction. Jones et al. identify recurring influence mechanisms, Armstrong et al. show how provisional honesty can change as a call develops, and Heiding et al. demonstrate that contemporary AI-generated speech can operate within persuasive scam scenarios at substantial scale. None of these findings implies that identity acceptance, continued interaction, and security compromise are equivalent outcomes.

3.3. Research Gap and Positioning

Broader reviews show why these findings are difficult to combine directly. Triantafyllopoulos et al. describe vishing as an interdisciplinary problem spanning deception, persuasion, speech generation, detection, and intervention, while identifying a lack of speech-technology research and suitable vishing data that connect these areas [3]. At the broader social-engineering level, Burda, Allodi, and Zannone systematically reviewed 169 empirical studies covering 735 hypotheses and found that experimental research only partially represents the attack space found in realistic social engineering [14]. Context, cognitive processes, variation in pretexts, and different degrees of targetization are among the factors that remain incompletely represented. These findings caution against deriving a complete call-based threat model from either isolated acoustic experiments or individual

persuasion studies.

The reviewed literature consequently establishes the main components of the threat but leaves an important analytical separation unresolved. Voice-perception research can establish whether synthetic speech supports a target identity; vishing research can explain how credibility is maintained or challenged; and AI-vishing research can measure susceptibility to persuasive synthetic calls. The reviewed work does not, however, provide a call-specific security model that explicitly traces how an identity signal becomes relevant to a protected authorization decision while distinguishing the evidence supporting *who the caller appears to be* from the evidence required to authorize *what the caller requests*.

This thesis addresses that gap through a socio-technical analysis of call-based impersonation. Rather than treating a convincing clone or continued conversation as equivalent to a successful attack, the analysis separates identity success, interaction success, and security-impact success and examines the technical, contextual, human, and procedural conditions that connect them. The contribution is therefore not another measurement of whether synthetic speech sounds realistic, but an integration of existing evidence around the security transition from perceived caller identity to protected authorization.

4. Methodology

This chapter explains how the analytical framework was constructed, which elements are adopted from established research, which are adapted to the present problem, and which were developed specifically for this thesis. It also defines the logic used to select and compare the two scenarios, the limited role of the technical validation, and the evidential boundaries that constrain the conclusions. The purpose is methodological traceability: each analytical step should be attributable either to an external source or to an explicitly identified thesis-specific design decision.

4.1. Research Design and Evidence Strategy

The thesis uses a structured analytical design with four linked components. First, a targeted synthesis of literature establishes the technical, psychological, and security concepts needed for the analysis. Second, those concepts are integrated into a call-specific socio-technical threat model. Third, two constructed scenarios apply the same model under

deliberately contrasting relationship structures. Fourth, a bounded local technical validation tests selected implementation assumptions without being treated as a co-equal benchmark study. The primary contribution is therefore analytical: the technical validation constrains and contextualizes the model rather than defining its novelty.

The literature component is a targeted synthesis rather than a systematic review. Sources are selected according to their analytical role: peer-reviewed empirical studies support claims about perception and human behavior, primary technical publications support claims about speech systems, established methodological literature supports the modeling approach, and official guidance supports authentication and defensive-control analysis. Recent research is emphasized where technical capability or threat conditions are changing rapidly. Evidence is kept within the limits of the underlying study design; for example, perceived speech realism is not treated as evidence of target-speaker attribution, and identity attribution is not treated as evidence of authorization or attack success.

4.2. Construction and Provenance of the Analytical Framework

The analytical framework connects the attacker, the Decision-Maker, the protected process, and the conditions that determine whether a voice-based impersonation attempt can lead to security impact. It combines established security and behavioral concepts with structures developed specifically for this thesis.

The first element is the B1–B3 boundary model. It adapts the general security concept of trust boundaries to call-based impersonation [15]; the exact three-boundary decomposition is author-developed. B1 represents the presentation of attacker-controlled identity cues. B2 is the central decision boundary at which the DM determines whether caller-derived identity and plausibility are sufficient or whether the authorization evidence required by the protected process must be obtained. B3 represents the transition from that human decision into a protected organizational action. A failure at B2 occurs when caller-derived cues are allowed to substitute for the required authorization evidence. The boundaries are developed in detail in Chapter 5.

The second element is an attack tree, which represents the logical conditions required for the attack to succeed. The thesis uses the established AND/OR structure of attack trees [16], [17]. The specific root goal, branches, voice-related conditions, and verification dependency were developed for this thesis. The tree is used qualitatively: it shows which conditions must occur together or which alternatives may satisfy the same requirement, but it does not assign probabilities to them.

The third element is a seven-stage lifecycle, which shows how the attack develops over

time. It was created by ordering the main prerequisites and decision points identified in the threat model and attack tree. Although staged attack models already exist, the seven stages used here are not copied from a single framework. They were developed specifically for voice-cloning-based telephone impersonation and are not intended as a universal social-engineering lifecycle.

Truth-Default Theory (TDT) is used as the main behavioral framework for the interaction itself. Unlike the structures above, TDT is an established external theory. Armstrong et al. show in a vishing context that callers may initially be treated as truthful until sufficiently diagnostic information creates suspicion [4]. In this thesis, TDT is used to explain how provisional caller legitimacy can persist, how inconsistencies may trigger suspicion, and why verification can become important during the interaction. Finally, the thesis distinguishes three levels of success:

1. **Identity success:** the claimed caller identity is accepted as sufficiently plausible for the interaction to continue.
2. **Interaction success:** the caller maintains sufficient credibility through the decision-relevant exchange.
3. **Security-impact success:** a protected action occurs without the authorization evidence normally required for that action.

This distinction was developed for the thesis and prevents successful identity presentation or sustained interaction from being treated as equivalent to security compromise. The contribution lies in integrating these elements around one specific problem: how synthetic caller identity can influence a human decision before that decision reaches a protected authorization process.

4.3. Scenario Selection and Analysis

Two constructed scenarios are used to apply the analytical framework. They are selected for analytical contrast rather than as representative examples of all voice-cloning fraud. Both use the same basic structure: a telephone interaction, impersonation of a specific person known to the Decision-Maker (DM), a protected financial action, and an independent verification requirement. The main difference is the relationship through which the attacker attempts to influence the DM.

Prior familiarity is a deliberate condition in both scenarios. It gives the cloned voice an identity-specific function because the DM can associate it with a known person. The existing relationship can also change the interpersonal cost of refusing or delaying the request: research on social distance and prosocial behavior indicates that willingness to

assist generally increases with social closeness [18]. Familiarity is therefore modeled as both an identity cue and a possible source of social pressure, without assuming that it is sufficient to override formal procedure.

The two scenarios vary the form of this relationship. Scenario 1 uses a professional relationship in which familiarity is combined with executive authority. Scenario 2 removes hierarchy and instead combines a service relationship with limited social familiarity. This allows the analysis to compare whether different forms of interpersonal pressure can contribute to the same underlying security problem: allowing perceived caller legitimacy to influence a protected authorization decision.

The same eight analytical dimensions are applied to both scenarios:

protected process and decision authority; claimed identity and relationship to the DM; reference-audio conditions; identity-presentation conditions; dominant influence mechanism; factors maintaining or challenging perceived legitimacy; independent verification path; and success and failure conditions.

These dimensions were derived from the research questions and B1–B3 threat model: they cover the protected decision and actor context, technical prerequisites and identity presentation, interaction and influence, the independent authorization path, and success and failure conditions. They therefore provide a common structure for comparing the complete modeled attack path across both scenarios.

4.4. Supplementary Technical Validation

The supplementary technical validation examines selected technical assumptions of the threat model using XTTS-v2, RVC v2, and Seed-VC. The final scenario-linked evaluation measures linguistic preservation through WER/CER, speaker-embedding similarity through CAMPPlus, processing speed through RTF, and perceptual characteristics through a blinded single-author inspection. XTTS generates each of two fixed German scenario utterances three times, while RVC and Seed-VC convert the same six human source recordings. The configurations are fixed before evaluation and technically abnormal outputs are retained. The evaluation is used to identify technical boundary conditions rather than to rank the systems or estimate deception success; detailed configurations and measurement procedures are provided in Appendix A.

4.5. Ethical and Evidential Boundaries

No real attack is carried out as part of this thesis. The executive and customer scenarios are analytical constructions, and neither identity is cloned in the technical evaluation. No employee, customer, organization, or other third party is contacted or deceived.

All speech material used in the exploratory technical validation was provided with consent. The thesis does not test caller-ID manipulation, attempt to bypass real authentication systems, or conduct deceptive calls against human participants. The technical work is limited to local speech generation and analysis.

The thesis also distinguishes between different types of evidence. Published empirical research is used to support claims about human perception and behavior. Security standards and official guidance support statements about authentication and defensive controls. The constructed scenarios are used to analyze how these elements can interact within the modeled attack process. The exploratory technical validation supports only findings obtained under the tested configurations and material.

To support reproducibility of the exploratory technical validation in Chapter 7, the scripts, preprocessing steps, model versions, configurations, benchmark texts, timing records, and measurement procedures are retained. The original voice recordings and generated samples are not published. The technical procedure can therefore be documented and reproduced with other suitable speech material without releasing the original speaker data.

5. Threat Model and Attack-Stage Analysis

This chapter develops the central threat model of the thesis. Voice cloning is not treated as the security outcome itself. Instead, the model traces how synthetic target-speaker speech can contribute to a perceived caller identity, how that perception enters a human decision, and under which conditions the decision can affect a protected organizational process.

Three views are used for different purposes. The B1–B3 boundary model identifies the transitions between identity presentation, human authorization, and the protected process. The attack tree represents the logical conditions required for security impact. The

seven-stage lifecycle places these conditions in temporal order. Together, they separate a technically plausible impersonation from a successful security compromise.

5.1. Scope, Actors, and Protected Decision

The model contains four principal actors or entities. The Threat Actor (TA) is an external adversary attempting to influence a protected decision through voice-based impersonation. The Impersonated Identity (II) is the person whose identity is presented by the attacker. The Decision-Maker (DM) is an organizational actor who can influence or execute the relevant protected action. Organization X (OX) represents the organizational and technical environment in which that action is governed.

The model focuses on protected actions for which the DM has legitimate authority but the caller does not. Examples include approving a payment, reauthorizing a stopped transaction, releasing protected information, or changing an account or transaction state. The relevant adverse outcome is therefore defined generally as:

a protected action performed without the authorization evidence normally required for that action.

This definition is important because the attacker does not need to obtain the DM's technical privileges. The attack instead attempts to influence a person who already possesses legitimate authority.

The model assumes a human-operated telephone interaction in which the attacker presents the identity of a specific person known to the DM. The attacker has access to target-speaker material and a synthetic-speech capability and can place the call. The protected process normally provides some form of verification or approval that does not depend solely on the incoming voice claim. Direct system compromise, autonomous voice agents, video deepfakes, attacks against voice-biometric systems, and implementation of caller-ID manipulation remain outside the model.

The model is independent of a particular voice-cloning implementation. Text-to-speech (TTS) and voice conversion (VC) are alternative technical means of producing target-like speech. Their practical limitations are considered separately in the exploratory technical validation in Chapter 7.

5.2. Identity, Human Authorization, and the Protected Process

The model separates the path from an incoming voice claim to a security-relevant outcome into three boundaries. Each boundary represents a different question: who the caller appears to be, whether the request is sufficiently authorized, and whether the protected action is ultimately executed.

B1 - Identity-Presentation Boundary. B1 is crossed when attacker-controlled signals reach the Decision-Maker (DM). The cloned voice is the central identity cue in this thesis, but it may be supported by contextual information, caller metadata, timing, or environmental cues. These signals can make the claim that the caller is a particular known person more plausible. B1 therefore concerns *perceived caller identity*; crossing it does not mean that the caller has been formally authenticated or that the requested action is authorized.

B2 - Human Authorization Boundary. B2 is the central decision boundary. It is reached when the DM must decide whether the caller's request may proceed. The security-relevant question is whether the DM relies on the caller's apparent identity and plausibility, or requires the independent authorization evidence defined by the protected process. This substitution may become more plausible when the synthetic voice resembles a person already known to the DM, because the voice can reinforce an existing identity judgment and make the caller appear more legitimate. A failure at B2 occurs when caller-derived cues are allowed to substitute for that required evidence.

B3 - Protected-Process Boundary. B3 is crossed when the DM's decision is translated into an organizational state change, such as approving a payment, reauthorizing a transaction, releasing protected information, or modifying an account. The attack reaches material security impact only when the protected action is executed without the authorization evidence normally required for that action.

The three boundaries therefore describe a progression from *perceived identity* to *authorization decision* to *protected action*. Voice cloning acts primarily at B1 by strengthening the identity claim. Its security relevance emerges at B2 when that identity claim is allowed to substitute for or weaken the authorization evidence required by the protected process, while actual impact occurs only after B3 is crossed. This separation is consistent with the broader security principle that protected operations should depend on explicit authorization checks rather than on preceding trust signals [13].

5.3. Attack Tree: Conditions for Security Impact

The attack tree represents the logical conditions required to reach the adverse outcome. It uses the established AND/OR attack-tree formalism, in which a root objective is refined into conditions that are jointly required or alternative ways of satisfying a requirement [16]. The concrete root objective and branch structure below were developed for this thesis.

The root objective is defined as:

Cause a protected action without the required authorization evidence through voice-based impersonation.

Figure 1 shows the logical decomposition of this objective.

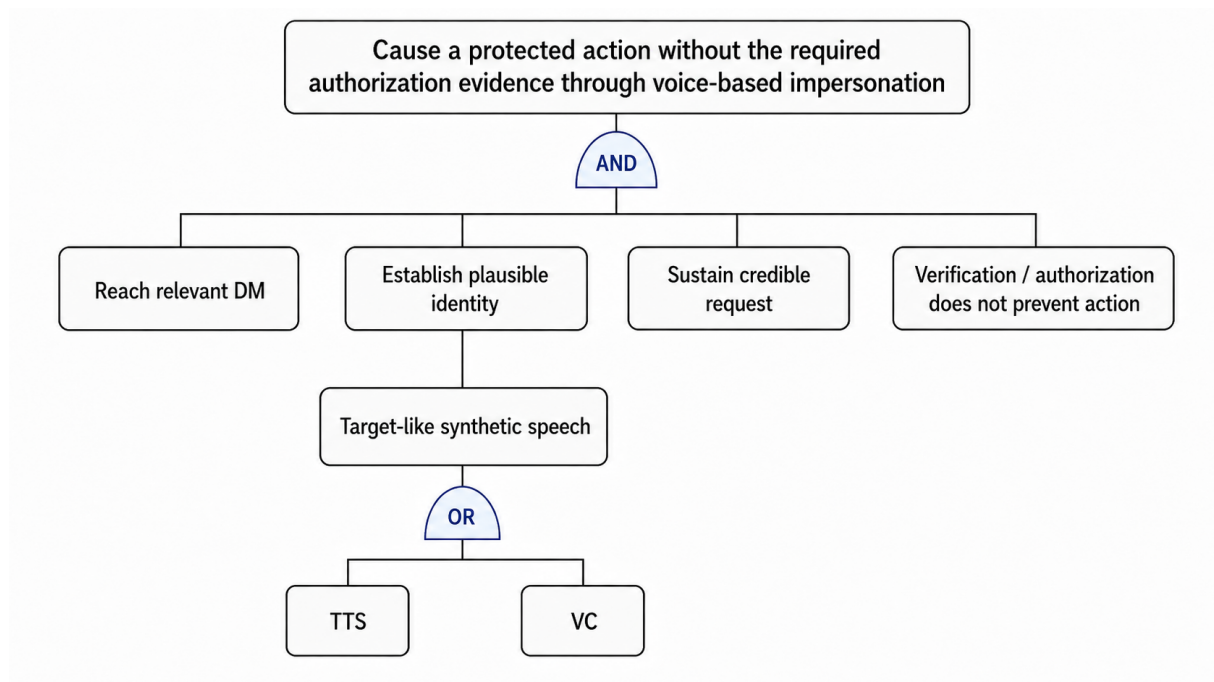


Fig. 1 Attack tree for a protected action performed without the required authorization evidence through voice-based impersonation. The concrete tree is author-developed using established AND/OR attack-tree semantics.

Four high-level conditions must align for this objective to be reached and therefore form an AND relationship:

1. A DM with relevant authority is reached.
2. A plausible target identity is established.
3. A credible request is sustained until the decision point.
4. Independent verification or authorization controls do not prevent the protected action.

The first condition connects target selection to actual security impact. A convincing impersonation directed at a recipient who cannot influence the protected process does not satisfy the root objective.

The second condition represents identity establishment. Suitable target-speaker material must be available and synthetic speech must reproduce enough characteristics of the target to support the identity claim in the interaction. TTS and voice conversion represent OR alternatives within this branch: either may supply the synthetic target-speaker component. Contextual information and other caller cues can reinforce the identity presentation but do not constitute authentication on their own.

The third condition concerns the interaction rather than the voice alone. The request, context, timing, linguistic content, and speech delivery must remain sufficiently coherent for the interaction to reach the protected decision. An identity presentation that is initially accepted can therefore still fail through contradictions, unstable speech output, unexpected questions, or other cues that generate sufficient suspicion.

The fourth condition connects social engineering to authorization. The protected outcome remains unreachable if an independent verification mechanism is correctly executed and enforced. From the attacker's perspective, this branch can be satisfied if verification is not initiated, is not completed adequately, or does not prevent the requested action. The model does not require the attacker to technically defeat the verification mechanism; influencing the process so that independent evidence is not required can be sufficient.

The attack tree also clarifies the three success levels introduced above. Establishing the identity can produce identity success. Sustaining that identity and request until the decision point can produce interaction success. The root objective is reached only when these conditions are combined with failure of the authorization barrier, producing security-impact success. The tree is used qualitatively and does not estimate attack probability. Its main implication is that target-speaker similarity alone is insufficient for security-impact success.

5.4. Seven-Stage Lifecycle

The attack tree describes what must be true for security impact to occur. The lifecycle describes when the relevant conditions arise. The following seven stages were developed for this thesis by ordering the prerequisites and decision points identified in the threat model. They are specific to the analyzed form of telephone impersonation and are not proposed as a universal social-engineering taxonomy.

Stage 1 - Target and Decision-Point Selection. The attacker identifies a protected process and a DM whose authority is relevant to the intended outcome. The target is therefore selected not only because the person can be contacted, but because their decision can influence the protected process.

Stage 2 - Context Acquisition. The attacker acquires information needed to make the claimed identity and request consistent with the circumstances of the call. Relevant information can concern roles, relationships, current events, timing, or the expected reason for contact. Context improves plausibility but does not authenticate the caller.

Stage 3 - Reference-Audio Acquisition and Assessment. Target-speaker audio is obtained and assessed for its usefulness in producing synthetic speech. The important distinction is between available audio and usable reference material. Quantity alone does not guarantee that the material captures the speech characteristics needed for the intended interaction.

Stage 4 - Interaction Preparation. The claimed identity, requested action, context, and expected course of the conversation are aligned before contact. The main requirement at this stage is coherence: the reason for the call and the requested action should not immediately conflict with information already available to the DM.

Stage 5 - Identity Presentation and Call Establishment. The call begins and attacker-controlled cues cross B1. Synthetic target-like speech interacts with the DM's prior knowledge of the impersonated person and the surrounding context. If the caller is accepted as sufficiently plausible for the exchange to continue, identity success has been achieved. No protected action has yet occurred.

Stage 6 - Credibility Maintenance, Influence, and Verification. The interaction progresses toward the decision point. The caller must maintain sufficient credibility while responding to the DM and supporting the request. Truth-Default Theory is relevant at this stage because an initially accepted caller may continue to be treated as legitimate until sufficiently diagnostic information produces suspicion [4].

Influence operates within this interaction but is not itself treated as authorization. Relationship-specific mechanisms may change how the DM perceives delay, refusal, or verification, while inconsistencies can have the opposite effect and trigger scrutiny. The specific mechanisms are examined in Chapter 6 rather than assumed to operate identically across scenarios.

Independent verification becomes decisive at this stage. If the DM moves from caller-

derived cues to evidence whose trust does not depend on the incoming call, the attacker must either satisfy that independent requirement or the attack path terminates. Maintaining sufficient credibility until this decision point constitutes interaction success.

Stage 7 - Authorization and Security Outcome. The request is verified, rejected, escalated, or translated into a protected action. Security-impact success occurs only if the protected process is altered without the authorization evidence normally required for that action.

This final stage establishes the difference between deceiving a person and compromising a protected process. A call can sound credible, be attributed to the intended identity, and continue for an extended period while still failing completely if independent authorization prevents the requested action.

Taken together, the boundary model, attack tree, and lifecycle show that voice cloning primarily strengthens the presentation of a claimed identity. Security impact requires additional conditions: the identity must remain credible until the relevant decision point, and the required verification or authorization process must fail to prevent the protected action. Chapter 6 applies this structure to two scenarios that differ in their social relationship and influence mechanism.

6. Scenario Analysis

This chapter applies the threat model from Chapter 5 to two constructed scenarios. Both scenarios involve a Decision-Maker (DM) who already knows the impersonated person's voice and has authority over a protected financial action. What differs is the relationship through which the attacker attempts to influence the decision. The first scenario relies primarily on professional hierarchy and authority; the second combines limited social familiarity with distress and a service relationship.

6.1. Scenario 1: Executive-Identity Impersonation

6.1.1. Scenario Setup

The first scenario concerns a payment-control process in Organization X. The DM is responsible for handling a payment that requires additional authorization before it can proceed. The Impersonated Identity is a senior executive with whom the DM has previously interacted through meetings, calls, or comparable professional communication. The DM therefore recognizes the executive's voice and has a general expectation of how the person communicates.

The Threat Actor (TA) presents as this executive and requests that the payment be released or approved despite the normal authorization requirement. Public appearances, interviews, social-media material, or other recordings provide potential reference audio for cloning the executive's voice. The scenario assumes that the attacker also possesses enough contextual information to make the timing and purpose of the call plausible.

The protected process itself is unchanged by the call. The payment normally requires written, secondary, or otherwise independent authorization. The attack therefore does not attempt to obtain the executive's technical credentials; it attempts to convince the DM to make an exception to the established process.

6.1.2. Identity and Influence

The dominant influence mechanism is authority. The caller does not merely ask for assistance but presents the payment as an executive instruction that should be carried out without further delay. The call occurs while the executive is supposedly travelling between business appointments. A hurried delivery and travel-related background noise are consistent with this explanation and help make the unusual direct call appear less exceptional.

The hierarchical relationship gives the request additional weight. Research on vishing identifies authority as a frequently used persuasion principle [5]. In this scenario, hierarchy is therefore modeled as increasing the perceived professional cost of delaying or challenging the request; it is not assumed to cause compliance by itself.

Urgency strengthens this pressure but is not treated as an independent cause of compliance. The caller presents the payment as time-sensitive and implies that the normal approval can be completed afterward. The resulting pressure comes from the combination of a recognizable executive identity, a plausible business context, and the perceived consequences of resisting a senior authority figure.

Truth-Default Theory helps explain why this combination can matter. If the caller has already been provisionally accepted as the executive, a coherent explanation and recognizable voice can allow that assumption to persist until a sufficiently strong inconsistency or the request to bypass procedure triggers suspicion [4].

6.1.3. Authorization Analysis

At B2, the DM must separate recognition of the executive from authorization of the request. The caller's apparent identity, seniority, and urgency may make refusal or delay appear professionally costly, but they do not replace the approval required for the payment. The security-relevant failure occurs if the DM treats the apparent executive instruction as sufficient reason to bypass that requirement.

A mandatory independent approval prevents this transition. The DM does not need to determine with certainty whether the voice is genuine; the request can instead be evaluated through the authorization process already required for the payment. A convincing voice may therefore produce identity success and even interaction success while still failing before security-impact success.

The main weakness exposed by the scenario is consequently not voice recognition itself, but discretionary authorization: the attack becomes possible when an apparently legitimate executive request can be treated as a reason to bypass the control designed to govern that same request.

6.2. Scenario 2: Retail-Customer Impersonation

6.2.1. Scenario Setup

The second scenario concerns a payment that has been stopped by Organization X Financial Services and requires customer authentication before it can be reauthorized. The DM works in transaction or fraud review and has authority to release the stopped transaction once the required authentication has been completed.

The Impersonated Identity is the affected account holder and is also a peripheral social acquaintance of the DM. They are not close friends, but the DM has encountered the person often enough to recognize the voice and know limited aspects of the person's personality and background. This relationship is independent of the DM's professional role.

The scenario assumes a targeted attack in which prior reconnaissance has established the relevant overlap between the customer, the DM, and the protected transaction. It

therefore has a higher information requirement than the executive scenario. The analysis does not assume that such a pairing is common; the overlap is treated as an explicit prerequisite of the constructed scenario.

The TA contacts the DM while presenting as the customer and requests reauthorization of the stopped payment. The normal process requires customer authentication before the DM may release it. The attack attempts to replace the required customer authentication with social recognition and a convincing explanation of the customer's circumstances.

6.2.2. Identity and Influence

The caller presents a specific urgent problem rather than distress in the abstract. The stopped transaction concerns the purchase of prescription medication, and the caller claims to be waiting at the pharmacy while being unable to complete the normal in-app authentication. The DM knows the impersonated customer as a peripheral acquaintance and recognizes the voice. The caller sounds increasingly overwhelmed and near tears as the interaction continues. The immediate difficulty of obtaining the medication provides the scenario's central source of distress.

The existing relationship changes the social meaning of the request, but familiarity is not assumed to benefit the attacker uniformly. It can reinforce an existing identity hypothesis while also giving the DM more speaker-specific expectations against which inconsistencies can be noticed [10]. Research on social distance supports the narrower proposition that socially closer relationships can increase prosocial behavior [18]; in this scenario, limited prior familiarity can therefore add interpersonal weight without being treated as sufficient to override authentication.

Distress adds a second source of pressure. Experimental research by Saulin et al. found that empathy toward another person's suffering can produce a prosocial decision bias [7]. The DM's professional role also points toward assistance: research on telephone service interactions in retail banking identifies helping and emotional support as elements of customer-oriented behavior [19]. In the scenario, these factors combine around one immediate problem: continuing to insist on authentication appears to prolong the difficulty of a distressed person whom the DM believes they recognize.

The attack therefore does not rely on distress alone. The recognizable voice supports the claimed identity, the existing acquaintance adds interpersonal weight, and the medication situation gives the request an immediate consequence. Together they create a pull toward assistance, while the required customer authentication remains unchanged.

6.2.3. Authorization Analysis

At B2, the DM faces a different form of pressure than in the executive scenario. The caller cannot invoke organizational authority. Instead, the interaction encourages the DM to view continued authentication as an obstacle to helping a distressed person whom the DM already recognizes.

The protected requirement nevertheless remains the same: recognition of the customer is not equivalent to completion of the required customer authentication. Security-impact success occurs only if the DM allows familiarity, apparent distress, or the plausibility of the voice to substitute for that authentication and then reauthorizes the transaction.

The strongest failure condition for the attacker is therefore straightforward enforcement of the existing process. If the customer must complete authentication through an independent channel before the DM can release the transaction, the social relationship does not need to be resolved or ignored. The DM may believe that the caller is genuinely the acquaintance and may still be unable to reauthorize the payment without the required evidence.

Scenario 2 therefore illustrates a pull toward assistance rather than pressure through hierarchy. The attacker attempts to make strict adherence appear unnecessarily harmful to someone the DM recognizes, while the security control requires the DM to separate willingness to help from authority to change the transaction state.

6.3. Cross-Scenario Comparison

The scenarios differ in how the existing relationship is used. In the executive scenario, the apparent identity places pressure on the DM through hierarchy: refusing or delaying the request can appear professionally costly. The interaction therefore pushes the DM toward compliance. In the customer scenario, the relationship works in the opposite direction. Recognition, distress, and the service role make assistance appear socially desirable, pulling the DM toward action.

Despite these different mechanisms, both scenarios reach the same authorization problem. The cloned voice supports attribution of the call to a person already known to the DM, while the relationship gives that apparent identity additional significance. Authority provides that significance in the executive scenario; familiarity and concern for the customer provide it in the second.

The common security failure occurs when this caller-derived legitimacy is allowed to substitute for independently required authorization evidence. The comparison therefore shows that different social pressures can converge on the same authorization weakness.

A verification or approval requirement whose trust does not depend on the incoming call can interrupt either path without requiring the DM to determine with certainty whether the voice is genuine.

7. Exploratory Technical Validation

This chapter tests selected technical assumptions of the threat model rather than benchmarking voice-cloning technology as a whole. The evaluation asks whether the tested systems can preserve German content, reproduce characteristics of a target speaker, retain scenario-relevant delivery, and operate at practically relevant processing speeds.

7.1. Evaluation Design

Three local systems were evaluated: XTTS-v2 [20] for text-to-speech (TTS), and RVC v2 [21] and Seed-VC [22] for speech-to-speech voice conversion (STS). All material was German. The workstation used an Intel Core i9-11950H, 64 GB RAM, and an NVIDIA RTX A5000 Laptop GPU with 16 GB VRAM. XTTS-v2 was evaluated as a zero-shot TTS system using a 22.8-second target reference, RVC-v2 used a trained target-speaker model based on approximately 31.6 minutes of prepared speech, and Seed-VC was evaluated in a zero-shot configuration. Full configurations, software versions, and retained evaluation artifacts are documented in Appendix A and the accompanying reproducibility repository.

All three systems targeted the same real speaker, although different genuine segments of that speaker were used as reference material. The references were taken from naturally spoken material rather than a controlled studio corpus, and their speaking rate and vocal presentation were not identical. Because reference selection can itself influence TTS and VC quality, reference-specific effects cannot be separated fully from model effects [8].

For the final scenario-linked test, XTTS generated each of the two scenario utterances three times. RVC and Seed-VC converted the same six human source recordings: three forceful executive takes and three distressed-customer takes. The final configurations were fixed before evaluating these outputs, and flagged outputs were retained rather than regenerated.

The evaluation combined four measures: WER/CER for linguistic preservation, CAMP-Plus cosine similarity for speaker-embedding similarity [23], real-time factor (RTF) for

processing speed, and a blinded single-author listening inspection. RTF is used only as a throughput measure for the respective pipeline and is not interpreted as a direct end-to-end latency comparison between TTS and STS.

The listening inspection rated intelligibility, scenario delivery, and naturalness on five-point scales. Target-speaker resemblance was evaluated separately in randomized listening sets using the corresponding genuine target-speaker reference material. An output was flagged when it exhibited a material content omission, substitution, repetition, abnormal duration, or truncation according to the predefined inspection criteria. The inspection is not a listener study and therefore provides descriptive observations rather than population-level perceptual evidence.

7.2. Technical Results

Table 1 summarizes the main objective results.

Tab. 1 Objective results of the scenario-linked technical validation.

System	Content	CAMPPlus	RTF	Flagged
XTTS-v2	5/6 WER = 0	0.72–0.77	0.40–0.43 ^a	1/6
RVC	mean WER 0.17	0.82–0.95	mean 0.38	2/6
Seed-VC	mean WER 0.36	0.90–0.94	mean 4.58	6/6

^aFive stable XTTS runs. One abnormal run had RTF 2.36.

CAMPPlus values are descriptive and are not interpreted as a direct cross-system ranking because the pipelines used different genuine reference segments of the target speaker. XTTS preserved the intended text exactly in five of six scenario generations. The remaining executive-scenario output stopped after the first sentence, deleting 8 of 22 words (WER 0.364; CER 0.404). The five successful generations operated at approximately RTF 0.40–0.43, showing that synthesis itself was faster than playback. This does not represent full conversational latency, because comprehension, response formulation, text entry, and playback are not included.

RVC preserved the source material considerably more reliably than Seed-VC and also operated faster than real time. Seed-VC showed substantially greater content instability, including omissions, substitutions, repetition, and abnormal duration. It was also slower than real time in the tested configuration. These results describe the specific configurations used here, not inherent limits of either architecture.

A particularly important result concerns CAMPPlus. RVC and Seed-VC produced substantial target-reference cosine similarities, and most STS outputs moved closer to the target embedding than the original source recording. However, this did not translate into convincing target identity during listening. Absolute embedding values are therefore not

treated as perceptual similarity scores or as directly comparable system rankings. Speaker embeddings and human speaker perception are related but non-equivalent measurements [24].

7.3. Blinded Perceptual Inspection

The blinded author ratings are shown in Table 2.

Tab. 2 Descriptive mean ratings from the blinded single-author inspection on a 1–5 scale ($n = 6$ outputs per system).

System	Intelligibility	Delivery	Naturalness	Resemblance
XTTS-v2	5.00	1.67	5.00	1.17
RVC	3.83	3.17	3.50	1.00
Seed-VC	2.67	1.00	2.17	1.00

XTTS illustrates the strongest separation between different quality dimensions. Its outputs were perceived as clear and natural, yet target-speaker resemblance and scenario-specific delivery remained weak. The executive utterances were generally too calm, while the distressed-customer condition conveyed only limited emotional intensity. Natural synthetic speech was therefore achievable without convincing target identity or appropriate scenario delivery.

RVC preserved scenario delivery better because the source speaker directly provided the intended forceful or distressed performance. Nevertheless, all six RVC outputs received a target-resemblance score of 1/5 despite CAMPPPlus values reaching approximately 0.95. Seed-VC likewise received 1/5 for target resemblance while also producing poorer naturalness and content stability.

The contrast is technically important: *content integrity, naturalness, speaker-embedding similarity, target-speaker resemblance, and prosodic delivery are distinct properties*. Strong performance on one did not imply strong performance on the others.

7.4. Interpretation and Boundary Conditions

Under the tested configurations, none of the three implementations reproduced the target speaker convincingly in the blinded author inspection. This result must not be interpreted as an upper bound on current voice-cloning capability.

The implementations were iteratively refined before the final evaluation, including reference selection, repeated synthesis testing, a larger RVC training run, and stabilization

of the final inference configurations. The reported results therefore represent final reproducible configurations selected after substantial technical refinement. Nevertheless, the evaluation remains limited to one target speaker and one STS source speaker, and different genuine segments of the same target speaker were used across the pipelines. Speaker- and reference-specific effects therefore cannot be separated completely from model effects.

The chosen systems also represent only a subset of available technology. Newer local architectures and carefully optimized open-source pipelines may perform differently, while controlled studies have reported substantially stronger perceptual results from some commercial systems than from XTTS-v2 [9], [25]. These studies use different speakers, languages, data, and evaluation conditions and therefore cannot be transferred directly to the present technical validation. They do, however, demonstrate that the weak local resemblance observed here is not evidence that contemporary voice cloning in general lacks convincing capability.

Taken together, the tested systems demonstrated several individual technical conditions relevant to the threat model: linguistic preservation, faster-than-playback processing in XTTS and RVC, preservation of source delivery in RVC, and measurable movement toward target-speaker embeddings. These properties did not, however, converge into convincing target-speaker impersonation in the blinded author inspection. The validation therefore shows that content preservation, naturalness, processing speed, speaker-embedding similarity, and perceived identity are distinct properties. The broader implications for identity judgment and authorization are discussed in Chapter 8.

8. Discussion and Conclusion

8.1. Answer to the Research Question and Main Finding

The main research question asks how voice cloning alters perceived caller legitimacy and under which conditions this identity signal can influence a protected authorization decision. The central finding is that voice cloning primarily strengthens the *identity claim*, not the caller’s authorization. A target-like voice can make it more plausible that a known person is calling, particularly when the voice is consistent with the expected identity, situation, and surrounding context. Security impact requires an additional transition: the

Decision-Maker (DM) must allow this perceived legitimacy to substitute for, weaken, or bypass the authorization evidence required by the protected process. Voice cloning is therefore security-relevant because it can strengthen an upstream identity judgment that later receives more decision weight than it should.

This distinction explains why a technically convincing clone is neither necessary nor sufficient for security-impact success. Research shows that listeners can have difficulty distinguishing high-quality synthetic voices from genuine speech [1], [26], while recent vishing research indicates that persuasiveness and scenario framing can be more important for compliance than voice human-likeness alone [2]. The relevant technical requirement is therefore not perfect reproduction of a speaker. The synthetic voice must be sufficiently consistent with the expected person to support the identity claim for long enough that the interaction reaches the protected decision. Conversely, even a highly convincing identity presentation can fail if the required authorization remains independent of the incoming call.

The scenario comparison sharpens this result by showing how the perceived identity becomes relevant through different social relationships. In the executive scenario, the caller combines a recognizable executive identity with hierarchical authority and urgency. The request is framed as a direct instruction, making procedural resistance professionally consequential for the DM: delaying or challenging the request means resisting a senior organizational authority. Authority is well established as a recurring persuasion mechanism in vishing [5].

The customer scenario produces pressure through a different mechanism. Here, recognition of a socially known customer is combined with credible distress and a time-sensitive personal need. This frames continued insistence on the normal procedure as withholding assistance from a person whom the DM believes they know and who is presented as being in immediate difficulty. Prior familiarity gives the request interpersonal relevance, while the DM’s professional service role provides an additional reason to assist. Research on social distance and empathy supports the narrower conclusion that existing relationships and perceived suffering can affect prosocial decision-making [7], [18].

The important result is that these different social mechanisms converge on the same authorization weakness. The attacker does not need authority and distress to operate in the same way. They only need the resulting legitimacy judgment to acquire inappropriate decision weight. In Scenario 1, the dangerous inference is effectively “this is the executive, therefore the exception should be made.” In Scenario 2, it is “this is the customer I know and they clearly need help, therefore the normal authentication can be relaxed.” In both cases, the security failure is the same: evidence supporting *who the caller appears to be* is treated as sufficient evidence that the requested protected action may proceed. This convergence also confirms why identity success must remain separate from security-impact

success: correctly attributing the caller to the claimed person does not itself justify the requested protected action.

The exploratory technical validation in Chapter 7 reinforces this distinction by showing that linguistic preservation, naturalness, processing speed, scenario-relevant delivery, speaker-embedding similarity, and perceived target identity can diverge substantially. XTTS-v2 produced clear and natural speech with largely preserved linguistic content and faster-than-playback synthesis, yet target-speaker resemblance and scenario-specific delivery remained weak in the blinded author inspection. RVC preserved the intended delivery more successfully and operated faster than real time, while its high CAMPPPlus similarities did not translate into convincing perceived target identity. Seed-VC likewise produced high target-reference embedding similarities but showed substantially weaker linguistic stability, naturalness, and processing speed in the tested configuration. The technical result is therefore not a ranking of systems, but evidence that no single technical measure establishes successful impersonation. These properties affect whether synthetic speech can sustain identity and interaction success, while security-impact success remains dependent on the separate authorization decision.

8.2. Security Implications and Transferability

The main defensive implication follows directly from this separation. A protected process should not require the DM to correctly determine whether an incoming voice is genuine in order to remain secure. Deepfake awareness and detection may create suspicion, but their effectiveness still depends on correctly interpreting the incoming audio. Independent authorization has a different security property. In Scenario 1, the payment remains protected when approval must arrive through a path independent of the executive’s call; in Scenario 2, the transaction remains protected when customer authentication must be completed independently of the customer’s call. In both cases, the evidence authorizing the action derives its trust from outside the interaction attempting to influence the DM. This principle is consistent with established authorization design and current NIST guidance on voice-based authentication [12], [13].

Reported incidents demonstrate why this distinction has practical relevance. In a widely reported 2019 case, an executive transferred approximately EUR 220,000 after a caller reportedly reproduced characteristics of a superior’s voice and presented an urgent payment request [27]. Such incidents do not establish how frequently the attack succeeds and cannot validate the constructed scenarios quantitatively. They do, however, demonstrate that synthetic or imitated identity, prior familiarity, organizational authority, urgency, and access to a financial decision can occur in the same attack context.

The distinction also transfers to private and family impersonation, although the authorization structure is different. A family member receiving an emergency call may recognize the apparent voice of a child, sibling, or other relative and also possess direct authority to transfer the requested money. B2 and B3 therefore remain conceptually distinct, but there may be little institutional separation between them: the same person judges the identity, decides whether the request is legitimate, and performs the payment. FTC and FBI guidance consequently recommends verification through information or channels that do not originate from the suspicious call, such as calling the person back using a known number or using a pre-arranged family phrase [28], [29]. These measures reproduce the central principle of the organizational model in a simpler form: recognition of the voice is separated from evidence that the request is genuine.

8.3. Limitations and Future Work

The analysis establishes a threat structure, not a probability of successful deception. The two scenarios are constructed analytical cases, and no participants were exposed to the modeled calls. The thesis can therefore identify the conditions under which voice cloning could become authorization-relevant and compare how different influence mechanisms interact with those conditions, but it cannot determine how often a real employee would accept the identity, remain engaged, or bypass the required control. The three success levels should consequently be understood as analytical distinctions rather than measured probabilities.

The scenario conclusions are also bounded by deliberate assumptions. Both scenarios require the DM to have prior familiarity with the impersonated speaker because the thesis examines identity-specific voice cloning rather than merely human-sounding synthetic speech. This strengthens the analytical relevance of cloning but prevents conclusions about recipients who have never heard the impersonated person. Familiarity is also not uniformly advantageous to the attacker: greater knowledge of a person’s voice, vocabulary, and behavior can provide additional mismatch cues [10]. Scenario 2 is narrower still because reconnaissance must identify an overlap between the DM’s social environment, the institution’s customer base, and a relevant transaction. The scenario therefore demonstrates the consequences once these conditions exist; it does not establish that identifying such a target combination is common or easy. Similarly, the push–pull distinction is derived from the two analyzed scenarios and should not be interpreted as a general taxonomy of social-engineering attacks.

The technical findings are bounded by the evaluation design. Chapter 7 examined three local German-language implementations using one target speaker and one STS source speaker, with different genuine target-speaker segments used across the pipelines. The

blinded perceptual inspection was conducted by a single evaluator, and no population listener study, live deception experiment, or controlled cross-language comparison was performed. The results therefore characterize the tested configurations and cannot establish general model performance, population-level speaker recognition, or deception probability. Future work should first extend the technical evaluation across multiple target and source speakers, standardized reference conditions, independent listeners, and additional contemporary systems. A subsequent human-subject study could then connect these technical properties to the analytical model by measuring identity acceptance, continued interaction, and authorization behavior separately. Such a design would directly test whether the three success levels identified in this thesis are empirically separable and how familiarity, authority, distress, and independent verification affect transitions between them.

8.4. Conclusion

Voice cloning weakens the assumption that a familiar voice is reliable evidence of caller identity, but a false identity judgment does not by itself create a security compromise. The threat becomes consequential when synthetic identity presentation survives the interaction, acquires social meaning through the relationship between caller and DM, and is allowed to influence a protected authorization decision.

The central security conclusion is therefore structural: *identity evidence and authorization evidence must remain separate*. A recognizable voice, plausible context, authority, familiarity, urgency, or distress may all make a caller appear legitimate, but none of these signals should be sufficient to authorize a protected action. If independently required authorization remains mandatory, the DM can be wrong about who is speaking without that error necessarily becoming security-impact success.

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Overview of AI technologies used

Technology	Use
OpenAI GPT and Google Gemini	Refinement of academic writing style. Generation of LaTeX syntax and formatting. Proofreading for grammatical, spelling, and punctuation errors. Explanation of complex reference papers for personal comprehension.
DeepSeek V4 Flash	Assistance with technical implementation and troubleshooting for the exploratory technical validation.

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A. Appendix

This appendix documents the information required to interpret and reproduce the exploratory technical validation in Chapter 7. Detailed scripts, configuration files, machine-readable results, and implementation-specific information are provided in the accompanying reproducibility repository.

A.1. Technical Environment and Final Configurations

All final tests were performed locally on the same workstation with an Intel Core i9-11950H processor, 64 GB RAM, and an NVIDIA RTX A5000 Laptop GPU with 16 GB VRAM. All reference, source, and generated speech used in the final evaluation was German. The three pipelines targeted the same real speaker but used different genuine segments of that speaker as reference material.

Tab. 3 Final system configurations used in the technical validation.

System	Configuration
XTTS-v2	Coqui TTS 0.22.0; <code>tts_models/multilingual/multi-dataset/xtts_v2</code> ; zero-shot TTS; German; fixed 22.8 s target-speaker reference; 24 kHz mono output.
RVC v2	Repository revision <code>81eed5e8f68b6bed1789f682fe78cdd324495afc</code> ; 40 kHz; approximately 31.59 min prepared target-speaker training speech; 800 epochs; RMVPE during preparation/training; Harvest for final inference; pitch shift +2; FAISS index; index rate 0.75.
Seed-VC	Repository revision <code>51383efd921027683c89e5348211d93ff12ac2a8</code> ; zero-shot voice conversion; approximately 23 s target-speaker reference; 22.05 kHz configuration; BigVGAN-v2 vocoder.

The complete inference parameters, software dependencies, local patches, and machine-readable configuration files are retained in the reproducibility repository. Model weights and genuine speaker recordings are not redistributed.

A.2. Evaluation Material

The final evaluation used two fixed German utterances derived from the scenarios in Chapter 6:

Tab. 4 Scenario utterances used in the final technical evaluation.

Scenario	Utterance
Executive	<i>Ich bin gerade zwischen zwei Terminen und die Zahlung muss heute noch bearbeitet werden. Die schriftliche Bestätigung reiche ich direkt danach nach.</i>
Customer	<i>Ich stehe gerade in der Apotheke und die Bestätigung in der App funktioniert nicht. Bitte helfen Sie mir mit der gestoppten Zahlung.</i>

XTTS-v2 generated each utterance three times, resulting in six outputs. For RVC and Seed-VC, three human performances of each utterance were recorded, producing six source recordings that were converted by both systems. The executive takes used a forceful and impatient delivery, while the customer takes used a distressed and overwhelmed delivery. Final configurations were fixed before evaluation, and technically abnormal outputs were retained rather than replaced.

A.3. Measurement and Perceptual Inspection

Four dimensions were evaluated independently: linguistic preservation, speaker-embedding similarity, processing speed, and perceptual characteristics.

Word Error Rate (WER) was calculated as

$$\text{WER} = \frac{S + D + I}{N},$$

where S , D , and I denote substitutions, deletions, and insertions, and N is the number of words in the reference text. Character Error Rate (CER) applies the same principle at character level. The intended scenario utterance was used as the headline reference text; source transcriptions were retained for the STS pipelines so that deviations already present in the spoken source could be distinguished from changes introduced during conversion.

Speaker similarity was measured using CAMPPlus embeddings and cosine similarity. Because the pipelines used different genuine target-speaker segments, absolute CAMPPlus values are treated descriptively and are not interpreted as direct cross-system rankings or probabilities of human speaker identification.

Processing speed was expressed as real-time factor:

$$\text{RTF} = \frac{t_{\text{processing}}}{t_{\text{audio}}}.$$

An RTF below 1 indicates that the measured synthesis or conversion operation completed faster than the corresponding audio duration. Model loading, human response formulation, and playback interaction were excluded; RTF therefore represents processing throughput rather than end-to-end conversational latency.

An output was flagged when it exhibited a material linguistic omission, substitution or insertion, repetition, truncation, or abnormal duration. Flagged outputs remained part of the evaluation.

The blinded single-author inspection covered all 18 final outputs. Files were assigned randomized identifiers before scoring. Intelligibility, scenario delivery, and naturalness were evaluated in a combined randomized set, while target-speaker resemblance was rated separately using the corresponding genuine target-speaker reference material. The system mapping was revealed only after scoring.

The rating dimensions were defined as follows:

Tab. 5 Dimensions used in the blinded single-author inspection.

Dimension		Criterion
Intelligibility		Clarity and correctness of the produced speech.
Scenario delivery		Match between vocal delivery and the intended forceful-executive or distressed-customer condition.
Naturalness		Absence of obvious synthetic, metallic, unstable, or otherwise distracting artifacts.
Target	resem-	Perceived similarity between the generated output and the genuine target-speaker reference.
blance		

The five-point scales used the following anchors. Intelligibility ranged from 1 (major content or intelligibility failure) to 5 (completely clear and correct). Scenario delivery ranged from 1 (intended delivery absent) to 5 (strongly conveyed). Naturalness ranged from 1 (severe synthetic, metallic, or unstable artifacts) to 5 (little or no obvious artifact). Target resemblance ranged from 1 (clearly does not resemble the target speaker) to 5 (strong target-speaker resemblance). Chapter 7 reports the descriptive arithmetic mean across the six outputs of each system. The inspection used one evaluator and therefore does not constitute a Mean Opinion Score study or provide population-level estimates.

A.4. Sensitivity Check and Reproducibility

The six STS source recordings had integrated loudness values between approximately -49.5 and -40.4 LUFS. To assess whether the low recording level materially influenced the results, gain-only copies were adjusted toward -28 LUFS without compression, denoising, clipping, or duration changes and processed with the same fixed RVC and Seed-VC configurations.

Tab. 6 Input-level sensitivity check for the STS pipelines.

System	Original		Adjusted	
	WER	CAMPPlus	WER	CAMPPlus
RVC	0.167	0.887	0.205	0.885
Seed-VC	0.364	0.918	0.273	0.912

RVC changed little after level adjustment. Seed-VC showed lower mean WER and one severe repetition/duration failure was no longer present, while mean CAMPPlus similarity remained similar. The sensitivity check therefore does not support low source level as a general explanation for the RVC results or the measured speaker-embedding similarities, although it may have contributed to individual Seed-VC instability.

The accompanying repository contains the final scripts, configuration files, upstream repository revisions, text-normalization code, WER/CER and CAMPPlus calculations, scenario texts, machine-readable results, failure records, blinded-rating material, and relevant preprocessing and training information. Genuine target-speaker recordings, human source recordings, identifiable generated audio, and speaker-specific model weights are excluded from the public repository; stable filenames and cryptographic hashes are used where necessary for traceability.

Repository: https://github.com/AbdulS123/Final_Bachelor_Thesis/